

SUNNY ZHOU

Machine Learning Engineer ◇ U.S. Citizen ◇ Berkeley, CA

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EDUCATION

Brandeis University, Michtom School of Computer Science
Master of Science in Computational Linguistics

Aug 2023 – May 2025

University of California, Davis, College of Letters and Sciences
B.S. in Mathematical and Scientific Computation; B.A. in Linguistics

Sep 2019 – Jun 2023

SKILLS

- **Awards** - Eagle Scout
- **Programming Languages:** - Python - Java - C - R - MATLAB - HTML - TypeScript - CSS
- **Libraries** - numpy - pandas - scikit-learn - pytorch - tensorflow - transformers - langchain - vllm - Flask - FastAPI - streamlit - sqlalchemy - sqlite - tableau
- **Technical Skills** - K12 Education - Numerical and Real Analysis - Deep Learning - Model Training & Fine-Tuning (PEFT/LoRA) - Model Evaluation & Error Analysis - Dataset Construction & Curation - Prompt Engineering - RAG & Vector Embeddings - Statistical Analysis - Data Pipeline Development - System Design - Docker - API Development

PROJECTS

ZYC at SemEval-2026 Task 5: Application of BERT-based Contextual Embeddings Similarity for WSD 2026

- Designed and compared four novel transformer-based architectures for word sense disambiguation, exploring methods of capturing information with in narrative context.
- Improved model generalization by applying masked language modeling, gradual layer unfreezing, and targeted data augmentation, reducing overfitting on a small labeled dataset, achieving **57% accuracy** on test set.

Ambiguity and Bias: A Departure from Classical Evaluation – Master’s Thesis

2025

- Crafted and defined a novel dataset and evaluation metrics for a previously unmeasured task, probing semantic knowledge in **Llama-3.3**, **DeepSeek-R1**, **Gemma-3**, and **Mistral-3.1** — demonstrating experience designing benchmarks for novel ML problems.
- Ran systematic error analysis across all four models using **WordNet**-derived ground truth, surfacing consistent failure modes and quantifying performance gaps across word sense categories.

Production-Style RAG System for Code Repository Q&A

2025

- Built a full-stack retrieval system covering document ingestion, chunking, and **HNSW vector indexing** of source code and README files, enabling natural language queries over a codebase.
- Deployed a **Streamlit** chat interface backed by a quantized **Llama** model, taking the system from initial prototype to a fully functional, user-facing product.

Agentic LLM Tutoring System with Step-by-Step Reasoning

2025

- Shipped an end-to-end agentic tutoring system using **Llama-3.3** and a custom **LangChain** tool suite, with a **Flask** web interface serving MathJax-rendered step-by-step solutions.
- Increased solution accuracy through iterative prompt evaluation, identifying failure patterns and refining tool descriptions to prevent premature answer disclosure.

Speech Act Dataset from NPR Podcast Transcripts

2024

- Designed an annotation schema for a novel dialogue act classification task, scraping and processing multilingual NPR transcripts with **BeautifulSoup4** and authoring labeling guidelines used across multiple annotators.
- Validated dataset quality using **Fleiss Kappa** inter-annotator agreement scoring, running adjudication rounds to resolve label conflicts and produce a clean corpus for downstream training.

Logistic Regression and Naive Bayes Fine Tuning for Multi-label Article Keyword Classification (2023)

- Constructed a fine-tuning paradigm for logistic regression and naive Bayes models implemented through scikit-learn, optimizing for overall accuracy and resulting in an accuracy of 98%.
- Collected and utilized gold training data from Kaggle, vectorizing article descriptions using uni-, bi-, and trigram counts.

Parent and Children Language Function Word Production (2023) – Undergraduate Thesis

- Analyzed the function and content word production frequency of parents and children, grouped between age 3-72 months, from the CHILDES database, using pandas and R.
- Identified traditional development psychology milestones were reflected in the interaction of function word categories (e.g. definite and indefinite articles).

EXPERIENCE

Substitute Teacher Twin Rivers Unified School District, McClellan, CA Apr 2026 – Present

- Delivered instruction across seven grade levels and special education in a high-need public school district serving a diverse, multilingual, and economically disadvantaged student population in northern Sacramento County.
- Adapted lesson plans and classroom management strategies across elementary, middle school, and special education settings on short notice.

Teaching Assistant Brandeis University, Waltham, MA Aug 2023 – Dec 2024

Courses: NLP I, Numerical Methods and Big Data, Linear Algebra, Semantics

- Taught and explained deep learning, optimization, and NLP algorithms to graduate and undergraduate students across four courses, holding weekly office hours and recitations.
- Co-designed homework assignments, grading rubrics, and solution sets with faculty, contributing to curriculum development across ML and mathematics courses.

Undergraduate Research Assistant University of California, Davis, Davis, CA Mar 2021 – Jun 2023

Language, Logic, and Learning Lab (L3Lab)

- Designed and ran psycholinguistics experiments on pragmatic inference in children, co-authoring study procedures and presenting findings at the UC Davis Language Symposium.

Undergraduate Research Assistant University of California, Davis

Language and Literacy Lab (LEAD): Dr. Yuuko Uchikoshi-Tonkovich; Davis, CA September 2020 - September 2021

- Transcribed and translated English-Mandarin children and parent interactions in Child Language Data Exchange System (CHILDES).
- Analyzed data of teacher-children bilingual interactions in head start programs using Excel and R.
- Presented research abstract at the UC Davis Undergraduate Research Conference (*Topic: Classroom Language Environment Effects on Dual Language Children Development*).

Summer Basketball Coach WeBasketball

Berkeley, CA

October 2014 - July 2019, July 2022

- Coached fundamentals basketball for elementary and middle school students in classes and summer camps.
- Navigated inter-teammate relations by creating a collaborative environment, facilitating conflict resolution.